Partial derivatives of Doppler equation

Carl Tape and Bella Seppi

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In[1589]:=

```
Clear[t, Ft, FfactorA, Ffactor]
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In[1590]:=

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Clear[v0, tp0, l, f0, c, tp]
```

Main equations for the Doppler frequency (Meng and Ben-Zion, 2018, Eq. 3b).
Note that tp is the dependent variable and should be thought of a time in the spectrogram.
Note that FfactorA < 0, Ffactor < 1, and Ft > f0.

In[1591]:=

$$\texttt{t[v0\_, tp0\_, l\_, c\_, tp\_] :=} \frac{(\texttt{tp}-\texttt{tp0}) - \texttt{Sqrt}\left[(\texttt{tp}-\texttt{tp0})^2 - \left(1-\frac{\texttt{v0}^2}{\texttt{c}^2}\right)\left((\texttt{tp}-\texttt{tp0})^2 - \frac{\texttt{l}^2}{\texttt{c}^2}\right)\right]}{\left(1-\frac{\texttt{v0}^2}{\texttt{c}^2}\right)};$$

$$\texttt{FfactorA[v0\_, tp0\_, l\_, c\_, tp\_] :=} \frac{\texttt{v0}}{\texttt{c}} \frac{\texttt{v0 t[v0, tp0, l, c, tp]}}{\texttt{Sqrt}\left[\texttt{l}^2 + (\texttt{v0 t[v0, tp0, l, c, tp]})^2\right]}$$

```
Ffactor[v0_, tp0_, l_, c_, tp_] := 1 + FfactorA[v0, tp0, l, c, tp];
```

$$\texttt{Ft[v0\_, tp0\_, l\_, f0\_, c\_, tp\_] :=} \frac{\texttt{f0}}{\texttt{Ffactor[v0, tp0, l, c, tp]}};$$

Find the limiting values for Fapproach (higher frequency; flying toward) and Frecede (lower frequency; flying away).
(This takes a minute.)

In[1595]:=

```
Limit[Ft[v0, tp0, l, f0, c, tp], tp → -Infinity]
Limit[Ft[v0, tp0, l, f0, c, tp], tp → Infinity]
```

Out[1595]=

$$\frac{\texttt{f0}}{1 - \frac{(c^2-\texttt{v0}^2)\sqrt{\frac{c^4\,\texttt{v0}^2\left(1+\sqrt{\frac{\texttt{v0}^2}{c^2}}\right)^2}{(c^2-\texttt{v0}^2)^2}}}{c^3\left(1+\sqrt{\frac{\texttt{v0}^2}{c^2}}\right)}}$$

Out[1596]=

$$\frac{\texttt{f0}}{1 + \frac{(-c^2+\texttt{v0}^2)\sqrt{\frac{c^4\,\texttt{v0}^2\left(-1+\sqrt{\frac{\texttt{v0}^2}{c^2}}\right)^2}{(c^2-\texttt{v0}^2)^2}}}{c^3\left(-1+\sqrt{\frac{\texttt{v0}^2}{c^2}}\right)}}$$

In[1597]:=

$$\text{Fapproach}[\text{v0\_}, \text{f0\_}, \text{c\_}] := \frac{\text{f0}}{1 - \frac{\left(c^2 - \text{v0}^2\right)\sqrt{\frac{c^4\,\text{v0}^2\left(1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)^2}{\left(c^2-\text{v0}^2\right)^2}}}{c^3\left(1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)}}$$

In[1598]:=

$$\text{Frecede}[\text{v0\_}, \text{f0\_}, \text{c\_}] := \frac{\text{f0}}{1 + \frac{\left(-c^2 + \text{v0}^2\right)\sqrt{\frac{c^4\,\text{v0}^2\left(-1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)^2}{\left(c^2-\text{v0}^2\right)^2}}}{c^3\left(-1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)}}$$

In[1599]:=

```
Fmean[v0_, f0_, c_] := (Fapproach[v0, f0, c] + Frecede[v0, f0, c]) / 2
```

In[1600]:=

```
Frange[v0_, f0_, c_] := Fapproach[v0, f0, c] - Frecede[v0, f0, c]
```

In[1601]:=

```
FullSimplify[Frange[v0, f0, c]]
```

Out[1601]=

$$\frac{\text{f0}}{1 + \frac{\left(-c^2+\text{v0}^2\right)\sqrt{\frac{1}{\frac{1}{c^2}+\frac{1-2\sqrt{\frac{\text{v0}^2}{c^2}}}{\text{v0}^2}}}}{c^3\left(1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)}} - \frac{\text{f0}}{1 + \frac{\left(-c^2+\text{v0}^2\right)\sqrt{\frac{1}{\frac{1}{c^2}+\frac{1+2\sqrt{\frac{\text{v0}^2}{c^2}}}{\text{v0}^2}}}}{c^3\left(-1+\sqrt{\frac{\text{v0}^2}{c^2}}\right)}}$$

Let ε = v0/c. Re-write as a series in ε, then consider approximations.

In[1602]:=

```
expr = Frange[v0, f0, c];
exprSub = expr /. v0 → ε c;
FullSimplify[exprSub]
Series[exprSub, {ε, 0, 4}] // Normal // FullSimplify
Series[exprSub, {ε, 0, 2}] // Normal // FullSimplify
```

Out[1604]=

$$\text{f0}\left(\frac{c - c\sqrt{\varepsilon^2}}{c\left(-1+\sqrt{\varepsilon^2}\right) + \left(-1+\varepsilon^2\right)\sqrt{\frac{c^2\left(\varepsilon^2+\varepsilon^4-2\left(\varepsilon^2\right)^{3/2}\right)}{\left(-1+\varepsilon^2\right)^2}}} + \frac{c\left(1+\sqrt{\varepsilon^2}\right)}{c\left(1+\sqrt{\varepsilon^2}\right) + \left(-1+\varepsilon^2\right)\sqrt{\frac{c^2\left(\varepsilon^2+\varepsilon^4+2\left(\varepsilon^2\right)^{3/2}\right)}{\left(-1+\varepsilon^2\right)^2}}}\right)$$

Out[1605]=

$$\frac{2\,\text{f0}\,\sqrt{c^2\,\varepsilon^2}\,\left(1+\varepsilon^2\right)}{c}$$

Out[1606]=

$$\frac{2\,\text{f0}\,\sqrt{c^2\,\varepsilon^2}}{c}$$

In[1607]:=

$$\text{Frangeapprox2}[v0\_, f0\_, c\_] := 2\, f0 \left(\frac{v0}{c}\right)$$

In[1608]:=

$$\text{Frangeapprox4}[v0\_, f0\_, c\_] := \text{Frangeapprox2}[v0, f0, c] \left(1 + \left(\frac{v0}{c}\right)^2\right)$$

This function will reduce some of the later expressions:

In[1609]:=

$$G[v0\_, tp0\_, l\_, c\_, tp\_] := \sqrt{\frac{-l^2\, v0^2 + c^2 \left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}$$

Here’s full simplification of Ft[ ], though I don’t think it helps:

In[1610]:=

```
FullSimplify[Ft[v0, tp0, l, f0, c, tp]]
```

Out[1610]=

$$\frac{f0}{1 - \frac{c\, v0^2 \left(-tp + tp0 + \sqrt{\frac{-l^2 v0^2 + c^2 \left(l^2 + (tp - tp0)^2 v0^2\right)}{c^4}}\right)}{\left(c^2 - v0^2\right) \sqrt{l^2 + \frac{c^4\, v0^2 \left(-tp + tp0 + \sqrt{\frac{-l^2 v0^2 + c^2 \left(l^2 + (tp - tp0)^2 v0^2\right)}{c^4}}\right)^2}{\left(c^2 - v0^2\right)^2}}}}$$

Here’s the expression in terms of G[ ]

In[1611]:=

$$\frac{f0}{1 - \frac{c\, v0^2\, (-tp + tp0 + G[v0, tp0, l, c, tp])}{\left(c^2 - v0^2\right) \sqrt{l^2 + \frac{c^4\, v0^2\, (-tp + tp0 + G[v0, tp0, l, c, tp])^2}{\left(c^2 - v0^2\right)^2}}}}$$

Out[1611]=

$$\frac{f0}{1 - \frac{c\, v0^2 \left(-tp + tp0 + \sqrt{\frac{-l^2 v0^2 + c^2 \left(l^2 + (tp - tp0)^2 v0^2\right)}{c^4}}\right)}{\left(c^2 - v0^2\right) \sqrt{l^2 + \frac{c^4\, v0^2 \left(-tp + tp0 + \sqrt{\frac{-l^2 v0^2 + c^2 \left(l^2 + (tp - tp0)^2 v0^2\right)}{c^4}}\right)^2}{\left(c^2 - v0^2\right)^2}}}}$$

Partial derivative expressions, fully simplified.
These expressions take a few minutes to derive. They can be commented out, since we define explicit expressions.

Speed of source, v0

In[1612]:=

```
(* FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],v0]] *)
```

In[1613]:=

$$\text{partialv}[\text{v0\_}, \text{tp0\_}, \text{l\_}, \text{f0\_}, \text{c\_}, \text{tp\_}] := -\left(\left(\text{f0}\,\text{v0}\left(-2\,l^4\,\text{v0}^4 + l^2\,(\text{tp}-\text{tp0})^2\,\text{v0}^6 + c^6\,(\text{tp}-\text{tp0})\left(2\,l^2 + (\text{tp}-\text{tp0})^2\,\text{v0}^2\right) G[\text{v0}, \text{tp0}, l, c, \text{tp}] + c^2\left(4\,l^4\,\text{v0}^2 - (\text{tp}-\text{tp0})^4\,\text{v0}^6 + l^2\,(\text{tp}-\text{tp0})\,\text{v0}^4\,(5\,\text{tp} - 5\,\text{tp0} - 3\,G[\text{v0}, \text{tp0}, l, c, \text{tp}])\right) - c^4\left(2\,l^4 - 3\,(\text{tp}-\text{tp0})^3\,\text{v0}^4\,(-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}]) - l^2\,(\text{tp}-\text{tp0})\,\text{v0}^2\,(-6\,\text{tp} + 6\,\text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])\right)\right)\right) \Big/ \left(c\,(c-\text{v0})\,(c+\text{v0})\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\sqrt{l^2 + \frac{c^4\,\text{v0}^2\,(-\text{tp}+\text{tp0}+G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2-\text{v0}^2\right)^2}} \left(c\,(-\text{tp}+\text{tp0})\,\text{v0}^2 + c\,\text{v0}^2\,G[\text{v0}, \text{tp0}, l, c, \text{tp}] - c^2\sqrt{l^2 + \frac{c^4\,\text{v0}^2\,(-\text{tp}+\text{tp0}+G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2-\text{v0}^2\right)^2}} + \text{v0}^2\sqrt{l^2 + \frac{c^4\,\text{v0}^2\,(-\text{tp}+\text{tp0}+G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2-\text{v0}^2\right)^2}}\right)^2\right)\right)$$

Unshifted frequency, f0
(note: it does **not** depend on f0)

In[1614]:=

```
(* FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],f0]] *)
```

In[1615]:=

$$\text{partialf}[\text{v0\_}, \text{tp0\_}, \text{l\_}, \text{f0\_}, \text{c\_}, \text{tp\_}] := \frac{1}{1 - \frac{c\,\text{v0}^2\,(-\text{tp}+\text{tp0}+G[\text{v0},\text{tp0},l,c,\text{tp}])}{\left(c^2-\text{v0}^2\right)\sqrt{l^2+\frac{c^4\,\text{v0}^2\,(-\text{tp}+\text{tp0}+G[\text{v0},\text{tp0},l,c,\text{tp}])^2}{\left(c^2-\text{v0}^2\right)^2}}}}$$

Arrival time, tp0

In[1616]:=

```
(* FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],tp0]] *)
```

In[1617]:=

$$\text{partialtp}[\text{v0\_}, \text{tp0\_}, \text{l\_}, \text{f0\_}, \text{c\_}, \text{tp\_}] :=$$
$$\left(\text{f0}\, l^2\, (c - \text{v0})\, \text{v0}^2\, (c + \text{v0})\, \left((-\text{tp} + \text{tp0})\, \text{v0}^2 + c^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\right)\right) \Big/$$
$$\left(c\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}}\right.$$
$$\left(c\, (-\text{tp} + \text{tp0})\, \text{v0}^2 + c\, \text{v0}^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}] - \right.$$
$$c^2\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}} +$$
$$\left.\left.\text{v0}^2\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}}\right)^2\right)$$

Distance to object, l

In[1618]:=

```
(* FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],l]] *)
```

In[1619]:=

$$\text{partiall}[\text{v0\_}, \text{tp0\_}, \text{l\_}, \text{f0\_}, \text{c\_}, \text{tp\_}] :=$$
$$\left(\text{f0}\, l\, (\text{tp} - \text{tp0})\, (c - \text{v0})\, \text{v0}^2\, (c + \text{v0})\, \left((-\text{tp} + \text{tp0})\, \text{v0}^2 + c^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\right)\right) \Big/$$
$$\left(c\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}}\right.$$
$$\left(c\, (-\text{tp} + \text{tp0})\, \text{v0}^2 + c\, \text{v0}^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}] - \right.$$
$$c^2\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}} +$$
$$\left.\left.\text{v0}^2\, \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}}\right)^2\right)$$

Acoustic wavespeed, c

In[1620]:=

```
(* FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],c]] *)
```

In[1621]:=

```
partialc[v0_, tp0_, l_, f0_, c_, tp_] := (f0 v0^2
    (-2 l^4 v0^4 + 2 l^2 (tp - tp0)^2 v0^6 + c^6 (tp - tp0) (l^2 + (tp - tp0)^2 v0^2) G[v0, tp0, l, c, tp] +
      c^2 (4 l^4 v0^2 - (tp - tp0)^4 v0^6 + l^2 (tp - tp0) v0^4 (3 tp - 3 tp0 - 4 G[v0, tp0, l, c, tp])) -
      c^4 (l^2 + (tp - tp0)^2 v0^2) (2 l^2 - 3 (tp - tp0) v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp] ))))/
  (c^2 (c - v0) (c + v0) G[v0, tp0, l, c, tp] Sqrt[l^2 + (c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)/(c^2 - v0^2)^2]
    (c (-tp + tp0) v0^2 + c v0^2 G[v0, tp0, l, c, tp] -
      c^2 Sqrt[l^2 + (c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)/(c^2 - v0^2)^2] +
      v0^2 Sqrt[l^2 + (c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)/(c^2 - v0^2)^2])^2)
```

First derivative w.r.t. time (tp)

In[1622]:=

```
FullSimplify[D[Ft[v0, tp0, l, f0, c, tp], {tp, 1}]]
```

Out[1622]=

$$-\left(\left(f0\, l^2\, (c - v0)\, v0^2\, (c + v0)\left((-tp + tp0)\, v0^2 + c^2\sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}\right)\right)\Bigg/\left(c\sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}\sqrt{l^2 + \frac{c^4\, v0^2\left(-tp + tp0 + \sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}\right)^2}{\left(c^2 - v0^2\right)^2}}\left(c\,(-tp + tp0)\, v0^2 + c\, v0^2\sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}} - c^2\sqrt{l^2 + \frac{c^4\, v0^2\left(-tp + tp0 + \sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}\right)^2}{\left(c^2 - v0^2\right)^2}} + v0^2\sqrt{l^2 + \frac{c^4\, v0^2\left(-tp + tp0 + \sqrt{\frac{-l^2\, v0^2 + c^2\left(l^2 + (tp - tp0)^2\, v0^2\right)}{c^4}}\right)^2}{\left(c^2 - v0^2\right)^2}}\right)^2\right)\right)$$

In[1623]:=

$$\text{D1F}[\text{v0\_}, \text{tp0\_}, \text{l\_}, \text{f0\_}, \text{c\_}, \text{tp\_}] :=$$

$$-\left(\left(\text{f0}\, l^2\, (c - \text{v0})\, \text{v0}^2\, (c + \text{v0})\, \left((-\text{tp} + \text{tp0})\, \text{v0}^2 + c^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}]\right)\right) \Big/ \left(c\, G[\text{v0}, \text{tp0}, l, c, \text{tp}] \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}} \left(c\, (-\text{tp} + \text{tp0})\, \text{v0}^2 + c\, \text{v0}^2\, G[\text{v0}, \text{tp0}, l, c, \text{tp}] - c^2 \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}} + \text{v0}^2 \sqrt{l^2 + \frac{c^4\, \text{v0}^2\, (-\text{tp} + \text{tp0} + G[\text{v0}, \text{tp0}, l, c, \text{tp}])^2}{\left(c^2 - \text{v0}^2\right)^2}}\right)^2\right)\right)$$

Second derivative w.r.t. time (tp)

In[1624]:=

```
(*FullSimplify[D[Ft[v0,tp0,l,f0,c,tp],{tp,2}]]*)
(* this calculation did not finish in >12 hours *)
```

In[1625]:=

```
(* Simplify[D[Ft[v0,tp0,l,f0,c,tp],{tp,2}]] *)
```

In[1626]:=

```
D2f[v0_, tp0_, l_, f0_, c_, tp_] := (f0
  (2 c l^4 v0^4 (c^2 - v0^2)^4 G[v0, tp0, l, c, tp] (-tp v0^2 + tp0 v0^2 + c^2 G[v0, tp0, l, c, tp])^2 -
    1/c^2 v0^4 (-3 c^10 v0^2 G[v0, tp0, l, c, tp] (-tp + tp0 + G[v0, tp0, l, c, tp])^3
        (-tp v0^2 + tp0 v0^2 + c^2 G[v0, tp0, l, c, tp])^2 -
      c^4 l^2 v0^2 (-c^2 + v0^2) (-tp + tp0 + G[v0, tp0, l, c, tp])^2
        (l^2 (c^2 - v0^2)^2 + c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2) - 3 c^6 G[v0, tp0, l, c,
         tp] (tp - tp0 - G[v0, tp0, l, c, tp]) ((tp - tp0) v0^2 - c^2 G[v0, tp0, l, c, tp])^2
        (l^2 (c^2 - v0^2)^2 + c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2) +
      l^2 (-c^2 + v0^2) (l^2 (c^2 - v0^2)^2 + c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)^2)
     (-c v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp]) +
       (c^2 - v0^2) Sqrt[l^2 + (c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)/(c^2 - v0^2)^2]))) /
  (c^3 (c^2 - v0^2)^6 (G[v0, tp0, l, c, tp])^3 (l^2 + (c^4 v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2)/(c^2 - v0^2)^2)^3
    (1 - (c v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp]))/((c^2 - v0^2) Sqrt[l^2 + (c^4 v0^2 (-tp+tp0+G[v0,tp0,l,c,tp])^2)/(c^2-v0^2)^2]))^3)
```

In[1627]:=

```
(* Simplify[D[Ft[v0,tp0,l,f0,c,tp],{tp,3}]] *)
```

In[1628]:=

```
D3f[v0_, tp0_, l_, f0_, c_, tp_] :=
 -(((3 f0 (2 l^6 v0^6 (c^2 - v0^2)^6 (-l^2 v0^2 + c^2 (l^2 + (tp - tp0)^2 v0^2))
         (-tp v0^2 + tp0 v0^2 + c^2 G[v0, tp0, l, c, tp])^3 +
       2 c l^2 v0^6 (c^2 - v0^2)^2 G[v0, tp0, l, c, tp] ((tp - tp0) v0^2 - c^2 G[v0, tp0, l, c, tp])
        (-3 c^10 v0^2 G[v0, tp0, l, c, tp] (-tp + tp0 + G[v0, tp0, l, c, tp])^3
           (-tp v0^2 + tp0 v0^2 + c^2 G[v0, tp0, l, c, tp])^2 -
```

$$c^4\, l^2\, v0^2 \left(-c^2 + v0^2\right) (-tp + tp0 + G[v0, tp0, l, c, tp])^2 \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right) - 3\, c^6\, G[v0, tp0, l, c, tp] \\ (tp - tp0 - G[v0, tp0, l, c, tp]) \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right)^2 \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right) + \\ l^2 \left(-c^2 + v0^2\right) \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right)^2\Big)$$

$$\left(-c\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp]) + \left(c^2 - v0^2\right) \sqrt{l^2 + \frac{c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2}{\left(c^2 - v0^2\right)^2}}\right) +$$

$$\Big(5\, c^{10}\, v0^8 \left(-l^2\, v0^2 + c^2 \left(l^2 + (tp - tp0)^2\, v0^2\right)\right) (-tp + tp0 + G[v0, tp0, l, c, tp])^4 \\ \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right)^3 + \\ 3\, c^8\, l^2\, v0^8 \left(-c^2 + v0^2\right) G[v0, tp0, l, c, tp]\, (-tp + tp0 + G[v0, tp0, l, c, tp])^3 \\ \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right) \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right) - \\ 6\, c^6\, v0^6 \left(-l^2\, v0^2 + c^2 \left(l^2 + (tp - tp0)^2\, v0^2\right)\right) (-tp + tp0 + G[v0, tp0, l, c, tp])^2 \\ \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right)^3 \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right) + \\ c^4\, l^2\, (tp - tp0)\, v0^8 \left(-c^2 + v0^2\right) (-tp + tp0 + G[v0, tp0, l, c, tp])^2 \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right)^2 + \\ 3\, c^4\, l^2\, v0^6 \left(-c^2 + v0^2\right) G[v0, tp0, l, c, tp]\, (tp - tp0 - G[v0, tp0, l, c, tp]) \\ \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right) \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right)^2 + c^2\, v0^4 \\ \left(-l^2\, v0^2 + c^2 \left(l^2 + (tp - tp0)^2\, v0^2\right)\right) \left((tp - tp0)\, v0^2 - c^2\, G[v0, tp0, l, c, tp]\right)^3 \\ \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right)^2 - l^2\, (tp - tp0) \\ v0^6 \left(-c^2 + v0^2\right) \left(l^2 \left(c^2 - v0^2\right)^2 + c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2\right)^3\Big)$$

$$\left(c\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp]) + \left(-c^2 + v0^2\right) \sqrt{l^2 + \frac{c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2}{\left(c^2 - v0^2\right)^2}}\right)^2\Bigg)\Bigg) \Big/$$

$$\left(c^7 \left(c^2 - v0^2\right)^9 G[v0, tp0, l, c, tp]^5 \left(l^2 + \frac{c^4\, v0^2 (-tp + tp0 + G[v0, tp0, l, c, tp])^2}{\left(c^2 - v0^2\right)^2}\right)^{9/2}\right.$$

$$\left(-1+\frac{c\,\mathrm{v0}^2\,(-\mathrm{tp}+\mathrm{tp0}+G[\mathrm{v0},\mathrm{tp0},l,c,\mathrm{tp}])}{(c^2-\mathrm{v0}^2)\sqrt{l^2+\frac{c^4\,\mathrm{v0}^2\,(-\mathrm{tp}+\mathrm{tp0}+G[\mathrm{v0},\mathrm{tp0},l,c,\mathrm{tp}])^2}{(c^2-\mathrm{v0}^2)^2}}}\right)^4\Bigg)\Bigg)$$

Trying to define a series, to consider dropping higher-order terms.

In[1629]:=

```
expr = D2f[v0, tp0, l, f0, c, tp];
exprSub = expr /. v0 → ε c;
FullSimplify[Series[exprSub, {ε, 0, 6}]]
```

Out[1631]=

$$\frac{c^2\,\mathrm{f0}\left(2\,l^4+c\,(l^2)^{3/2}\left(-2\sqrt{\frac{l^2}{c^2}}+3\,\mathrm{tp}-3\,\mathrm{tp0}\right)\right)\varepsilon^4}{l^6}+\frac{1}{2\,l^4}3\,c^2\,\mathrm{f0}\left(-4\,l^2+4\,c^2\,(\mathrm{tp}-\mathrm{tp0})\left(3\sqrt{\frac{l^2}{c^2}}-2\,\mathrm{tp}+2\,\mathrm{tp0}\right)+\frac{c^3\,(\mathrm{tp}-\mathrm{tp0})^2\left(8\sqrt{\frac{l^2}{c^2}}-5\,\mathrm{tp}+5\,\mathrm{tp0}\right)}{\sqrt{l^2}}+c\sqrt{l^2}\left(4\sqrt{\frac{l^2}{c^2}}-7\,\mathrm{tp}+7\,\mathrm{tp0}\right)\right)\varepsilon^6+O[\varepsilon]^7$$

What is the difference between the picked F(tp0) and f0?

In[1632]:=

```
Ft[v0, tp0, l, f0, c, tp]
```

Out[1632]=

$$\frac{\mathrm{f0}}{1+\frac{\mathrm{v0}^2\left(\mathrm{tp}-\mathrm{tp0}-\sqrt{(\mathrm{tp}-\mathrm{tp0})^2-\left(-\frac{l^2}{c^2}+(\mathrm{tp}-\mathrm{tp0})^2\right)\left(1-\frac{\mathrm{v0}^2}{c^2}\right)}\right)}{c\left(1-\frac{\mathrm{v0}^2}{c^2}\right)\sqrt{l^2+\frac{\mathrm{v0}^2\left(\mathrm{tp}-\mathrm{tp0}-\sqrt{(\mathrm{tp}-\mathrm{tp0})^2-\left(-\frac{l^2}{c^2}+(\mathrm{tp}-\mathrm{tp0})^2\right)\left(1-\frac{\mathrm{v0}^2}{c^2}\right)}\right)^2}{\left(1-\frac{\mathrm{v0}^2}{c^2}\right)^2}}}}$$

In[1633]:=

```
Ft[v0, tp0, l, f0, c, tp0]
```

Out[1633]=

$$\frac{\mathrm{f0}}{1-\frac{\mathrm{v0}^2\sqrt{\frac{l^2\left(1-\frac{\mathrm{v0}^2}{c^2}\right)}{c^2}}}{c\left(1-\frac{\mathrm{v0}^2}{c^2}\right)\sqrt{l^2+\frac{l^2\,\mathrm{v0}^2}{c^2\left(1-\frac{\mathrm{v0}^2}{c^2}\right)}}}}$$

In[1634]:=

```
FullSimplify[Ft[v0, tp0, l, f0, c, tp0] - f0]
```

Out[1634]=

$$f0 \left(-1 + \frac{1}{1 - \frac{v0^2 \sqrt{\frac{l^2 (c-v0)(c+v0)}{c^4}} \sqrt{\frac{c^2 l^2}{c^2-v0^2}}}{c\, l^2}} \right)$$

This is just a formula in terms of f0, v0, and c:

In[1635]:=

$$\text{Simplify}\left[\frac{1}{1 - \frac{v0^2}{c^2}} - 1\right]$$

Out[1635]=

$$\frac{v0^2}{c^2 - v0^2}$$

In[1636]:=

$$\text{Fshiftfactor[v0\_, c\_]} := \frac{v0^2}{c^2 - v0^2}$$

In[1637]:=

```
Fshift[f0_, v0_, c_] := f0 Fshiftfactor[v0, c]
```

Example calculation:

In[1705]:=

```
vx = 121.6;
fx = 121.0;
cx = 311.1;
N[Ft[vx, tpx, lx, fx, cx, tpx] - fx]
N[Fshift[fx, vx, cx]]
N[Fshiftfactor[vx, cx]]
```

Out[1708]=

```
21.8201
```

Out[1709]=

```
21.8201
```

Out[1710]=

```
0.180331
```

Preliminaries

Find the slope of the doppler at the center.
That means finding the root for the second derivative.

In[1644]:=

```
(* Solve[D[Ft[v0,tp0,l,f0,c,tp],{tp,2}]==0,tp] *) (* not sure if this will run *)
```

Find the inflection points by setting the third derivative to zero and solving.

In[1645]:=

```
(* Solve[D[Ft[v0,tp0,l,f0,c,tp],{tp,3}]==0,tp] *)
(* this probably does not finish *)
```

Example plot

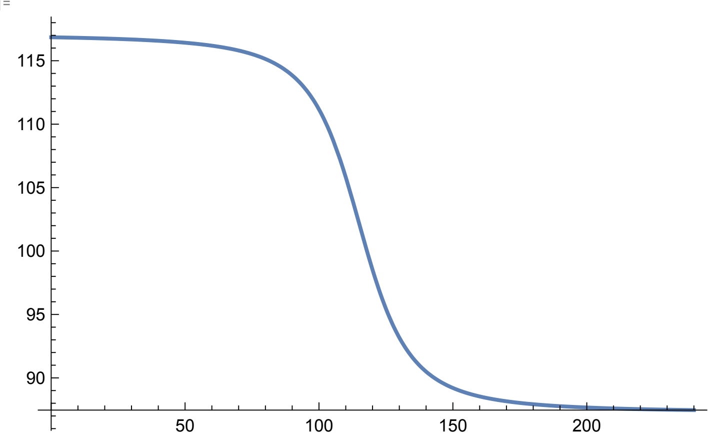
In[1646]:=

```
c = 343;
v0 = 50;
tp0 = 115;
l = 1000;
f0 = 100;
```

In[1651]:=

```
Plot[Ft[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

Out[1651]=



In[1652]:=

```
f0
```

Out[1652]=

```
100
```

In[1653]:=
```
N[Ft[v0, tp0, l, f0, c, tp0]]
N[FfactorA[v0, tp0, l, c, tp0]]
N[Ffactor[v0, tp0, l, c, tp0]]
```
Out[1653]=
```
102.171
```
Out[1654]=
```
-0.0212496
```
Out[1655]=
```
0.97875
```

list the approximations for the fmax (fminus) and fmin (fplus) frequencies

In[1656]:=
```
N[Fapproach[v0, f0, c]]
N[Frecede[v0, f0, c]]
N[Fmean[v0, f0, c]]
N[Frange[v0, f0, c]]
N[Frangeapprox4[v0, f0, c]]
N[Frangeapprox2[v0, f0, c]]
```
Out[1656]=
```
117.065
```
Out[1657]=
```
87.2774
```
Out[1658]=
```
102.171
```
Out[1659]=
```
29.7875
```
Out[1660]=
```
29.774
```
Out[1661]=
```
29.1545
```

Estimating a Doppler curve from data

1. Assume you know c, but maybe you’re oFf by a bit.

In[1662]:=
```
cX = 0.9 c;
```

2. Calculate fapproach and frecede.

In[1663]:=
```
FapproachX = 118.;
FrecedeX = 86.;
FrangeX = FapproachX - FrecedeX;
```

3. Calculate approximation to f0, then use it to get an approximate tp0. (So we pick the intersection of the horizontal line FtX with the spectrum.)

In[1666]:=
```
FtX = (FrecedeX + FapproachX) / 2
```
Out[1666]=
```
102.
```

In[1667]:=
```
tp0X = First[tp /. NSolve[Ft[v0, tp0, l, f0, c, tp] == FtX, tp]]
```
Out[1667]=
```
115.227
```

4. Get v0 using an approximation.

In[1668]:=
```
v0X = (cX FrangeX) / (2 FtX)
```
Out[1668]=
```
48.4235
```

Interlude: We now have curves with only two unknowns: tp and L.

In[1669]:=
```
FLtrue[L_, tp_] := Ft[v0, tp0, L, f0, c, tp]
FLest[L_, tp_] := Ft[v0X, tp0X, L, FtX, cX, tp]
```

In[1671]:=
```
N[FLtrue[L, tp]]
N[FLest[L, tp]]
```
Out[1671]=

$$\frac{100.}{1. + \frac{7.44687 \left(-115. - 1. \sqrt{-0.97875 \left(-8.49986\times10^{-6} L^2 + (-115. + tp)^2\right) + (-115. + tp)^2} + tp\right)}{\sqrt{L^2 + 2609.73 \left(-115. - 1. \sqrt{-0.97875 \left(-8.49986\times10^{-6} L^2 + (-115. + tp)^2\right) + (-115. + tp)^2} + tp\right)^2}}}$$

Out[1672]=

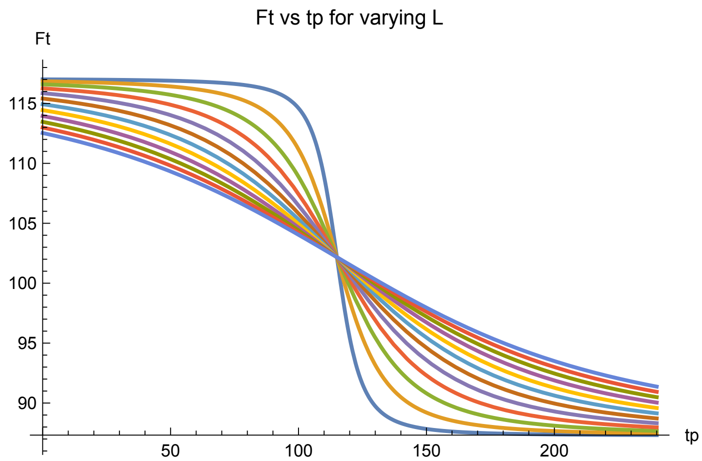
$$\frac{102.}{1. + \frac{7.78747 \left(-115.227 - 1. \sqrt{-0.975394 \left(-0.0000104937 L^2 + (-115.227 + tp)^2\right) + (-115.227 + tp)^2} + tp\right)}{\sqrt{L^2 + 2464.64 \left(-115.227 - 1. \sqrt{-0.975394 \left(-0.0000104937 L^2 + (-115.227 + tp)^2\right) + (-115.227 + tp)^2} + tp\right)^2}}}$$

Show how curves vary with changing L:

In[1673]:=

```
Plot[Evaluate@Table[FLtrue[L, tp], {L, 500, 6000, 500}], {tp, 0, 240},
 PlotRange → All, AxesLabel → {"tp", "Ft"}, PlotLabel → "Ft vs tp for varying L"]
```

Out[1673]=



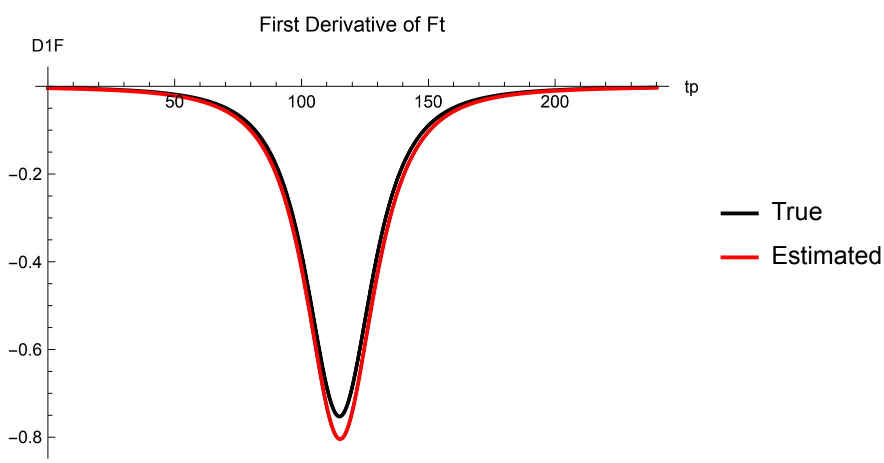
5. Calculate L from the slope of the Doppler at (t0p, f0).

Plot the slope (here we assume that l is known):

In[1674]:=

```
Plot[{
  Style[D1F[v0, tp0, l, f0, c, tp], Black],
  Style[D1F[v0X, tp0X, l, FtX, cX, tp], Red]},
 {tp, 0, 240}, PlotLegends → {"True", "Estimated"},
 PlotLabel → "First Derivative of Ft", AxesLabel → {"tp", "D1F"}, PlotRange → All]
```

Out[1674]=



Measure the slope by choosing two points on the linear section, then solve for

L.
KEY POINT: tp is set equal to tp0, so we are looking at the peak slope at the center of the Doppler curve.

In[1675]:=

```
FullSimplify[D1F[v0z, tp0z, lz, f0z, cz, tp0z]]
```

Out[1675]=

$$\frac{cz^3\ f0z\ v0z^2}{\frac{2\ cz\ lz^2\ v0z^2}{\sqrt{\frac{lz^2\ (cz-v0z)\ (cz+v0z)}{cz^4}}} - cz^4\sqrt{\frac{cz^2\ lz^2}{cz^2-v0z^2}} - v0z^4\sqrt{\frac{cz^2\ lz^2}{cz^2-v0z^2}}}$$

Cleaner expression, which factors out 1/L

In[1676]:=

$$\text{vfactor}[v0\_,\ c\_] := \frac{v0^2}{c}\left(1-\left(\frac{v0}{c}\right)^2\right)^{-3/2}$$

In[1677]:=

```
Lfactor[v0_, f0_, c_] := -f0 vfactor[v0, c]
```

In[1678]:=

$$\text{D1Forigin}[v0\_,\ l\_,\ f0\_,\ c\_] := \frac{\text{Lfactor}[v0,\ f0,\ c]}{l}$$

check that the “full simplify” expression reduces to the very clean version in D1Forigin:

In[1679]:=

```
D1Forigin[v0X, L, FtX, cX]
Simplify[D1F[v0X, tp0X, L, FtX, cX, tp0X]]
```

Out[1679]=

$$-\frac{804.278}{L}$$

Out[1680]=

$$-\frac{804.278}{\sqrt{L^2}}$$

solve for L:

In[1681]:=

```
xslope = -0.8;
Lfactor[v0X, FtX, cX] / xslope
```

Out[1682]=

```
1005.35
```

In[1683]:=
```
f0
```
Out[1683]=
```
100
```
In[1684]:=
```
N[vfactor[v0, c]]
```
Out[1684]=
```
7.52728
```

a numerical solve will produce + and - values:

In[1685]:=
```
Solve[D1F[v0X, tp0X, L, FtX, cX, tp0X] == xslope, L]
```
Out[1685]=
```
{{L → -1005.35}, {L → 1005.35}}
```

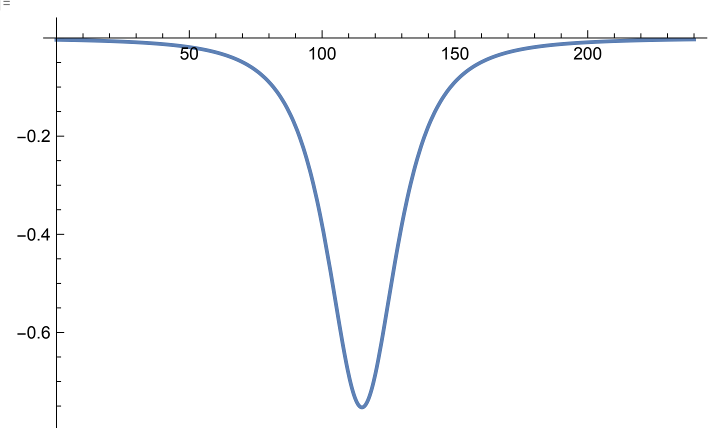
The number is slightly different if you use the measured slope and the **true** spectrum:

In[1686]:=
```
Solve[D1F[v0, tp0, L, f0, c, tp0] == xslope, L]
Lfactor[v0, f0, c] / xslope
```
Out[1686]=
```
{{L → -940.91}, {L → 940.91}}
```
Out[1687]=
```
940.91
```

Derivatives w.r.t. time

Note: the roots of the third derivative give time values at the inflection points.

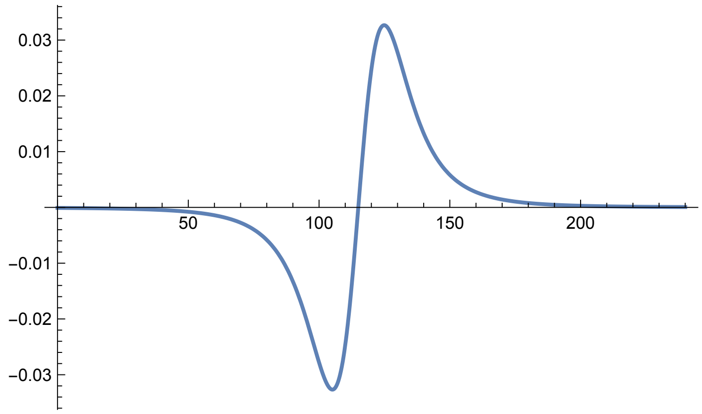
In[1688]:=
```
Plot[D1F[v0, tp0, l, f0, c, tp], {tp, 0, 240}, PlotRange → All]
```
Out[1688]=



In[1689]:=

```
Plot[D2f[v0, tp0, l, f0, c, tp], {tp, 0, 240}, PlotRange → All]
```

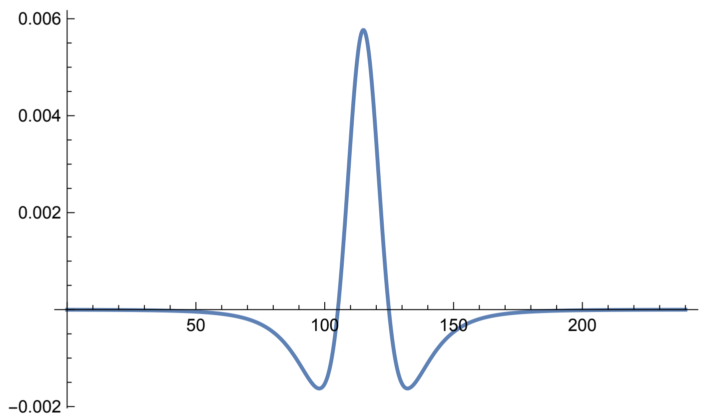
Out[1689]=



In[1690]:=

```
Plot[D3f[v0, tp0, l, f0, c, tp], {tp, 0, 240}, PlotRange → All]
```

Out[1690]=

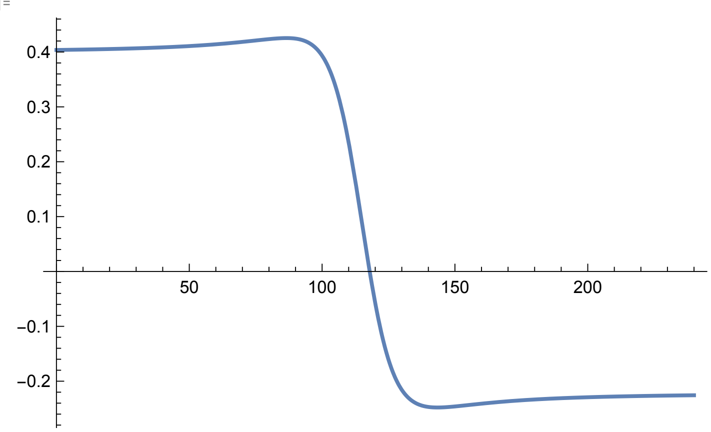


Partial derivatives for v0, tp0, l, f0, c

In[1691]:=

```
Plot[partialv[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

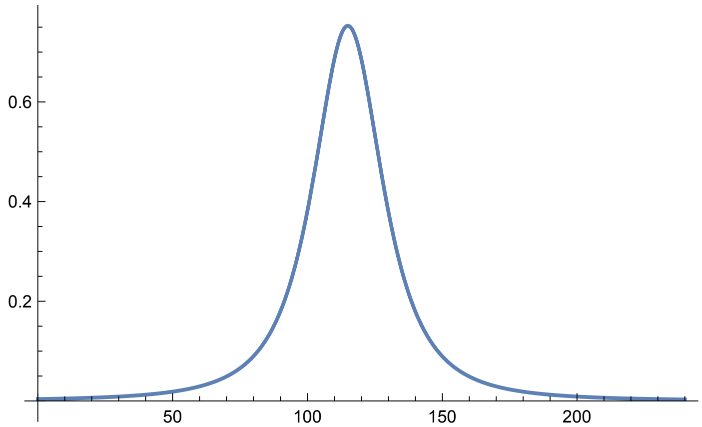
Out[1691]=



In[1692]:=

```
Plot[partialtp[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

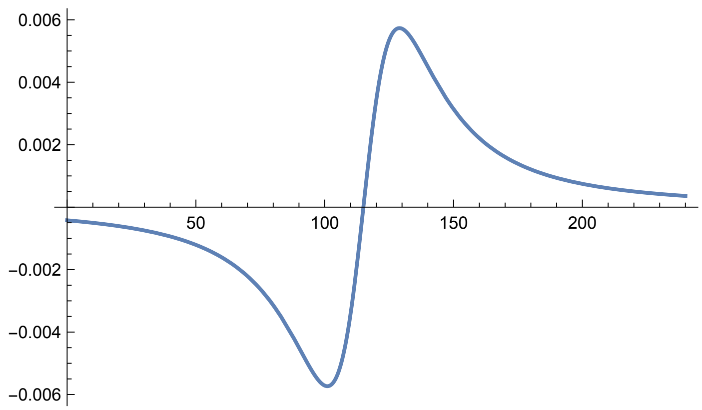
Out[1692]=



In[1693]:=

```
Plot[partiall[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

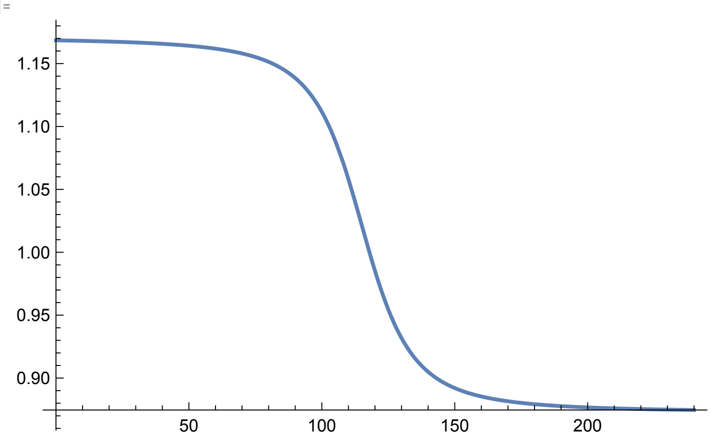
Out[1693]=



In[1694]:=

```
Plot[partialf[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

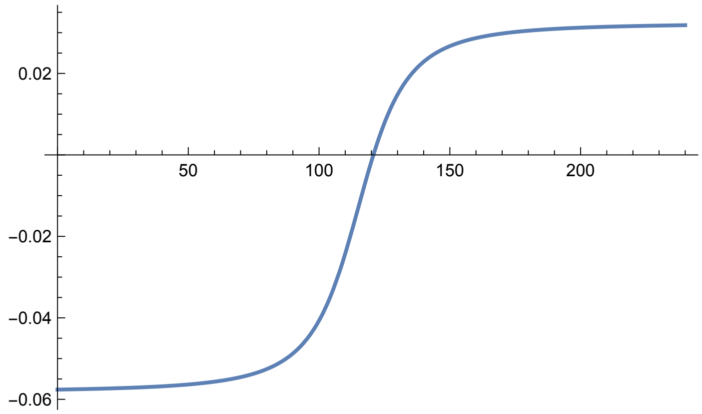
Out[1694]=



In[1695]:=

```
Plot[partialc[v0, tp0, l, f0, c, tp], {tp, 0, 240}]
```

Out[1695]=



Test values (for checking Python)

In[1696]:=

```
tp = 100;
```

In[1697]:=

```
N[t[v0, tp0, l, c, tp]]
N[Ft[v0, tp0, l, f0, c, tp]]
N[G[v0, tp0, l, c, tp]]
N[partialv[v0, tp0, l, f0, c, tp]]
N[partialtp[v0, tp0, l, f0, c, tp]]
N[partiall[v0, tp0, l, f0, c, tp]]
N[partialf[v0, tp0, l, f0, c, tp]]
N[partialc[v0, tp0, l, f0, c, tp]]
```

Out[1697]=

-19.0237

Out[1698]=

111.169

Out[1699]=

3.61945

Out[1700]=

0.393254

Out[1701]=

0.380922

Out[1702]=

-0.00571383

Out[1703]=

1.11169

Out[1704]=

-0.0406672